EMPIRICAL PROOF & MATHEMATICAL VERIFICATION
Thermodynamic Spectral Market Engine

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1. ARCHITECTURE & PHYSICS PROOF
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This framework completely abandons standard portfolio theory and neural
network black-boxes. It relies strictly on deterministic signal processing
and Non-Linear Topological Dynamics to track market capital flow.

MATHEMATICAL ENGINES USED:
a) 3rd-Order Tensor Matrix: Computes exact structural covariance coupled
   with volume-derivative friction to isolate real capital migration.
b) Spectral Graph Laplacian (Fiedler Value): Extracts the algebraic
   connectivity of the matrix (lambda_2) to dynamically dampen Kelly
   criterion sizing, shrinking bets when the market becomes hyper-fragile.
c) Kuramoto Phase Synchronization: Uses Hilbert Transforms to track the
   global phase oscillator r(t). It mathematically predicts systemic
   vaporization (crashes) when asset phases lock together (r > 0.80),
   triggering automated rotation into safe havens (TLT/GLD).

2. VERIFICATION METHODOLOGY
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To ensure complete clearance and absolute lack of look-ahead bias:
* IN-SAMPLE TRAINING (2008-2018): Used strictly to calibrate the
  Kuramoto crash threshold (0.80).
* OUT-OF-SAMPLE TESTING (2019-2024): Simulated completely blind. The
  engine ran forward with locked static parameters, executing a
  fractional Kelly allocation over the dynamic tensor matrix every 14 days.

3. INPUT & OUTPUT DATA STRUCTURE
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INPUT:  Continuous Time-Series OHLCV Ticks (yfinance).
OUTPUT: A vector of dimension N containing exact localized portfolio
        weights dynamically scaled by safety algorithms.
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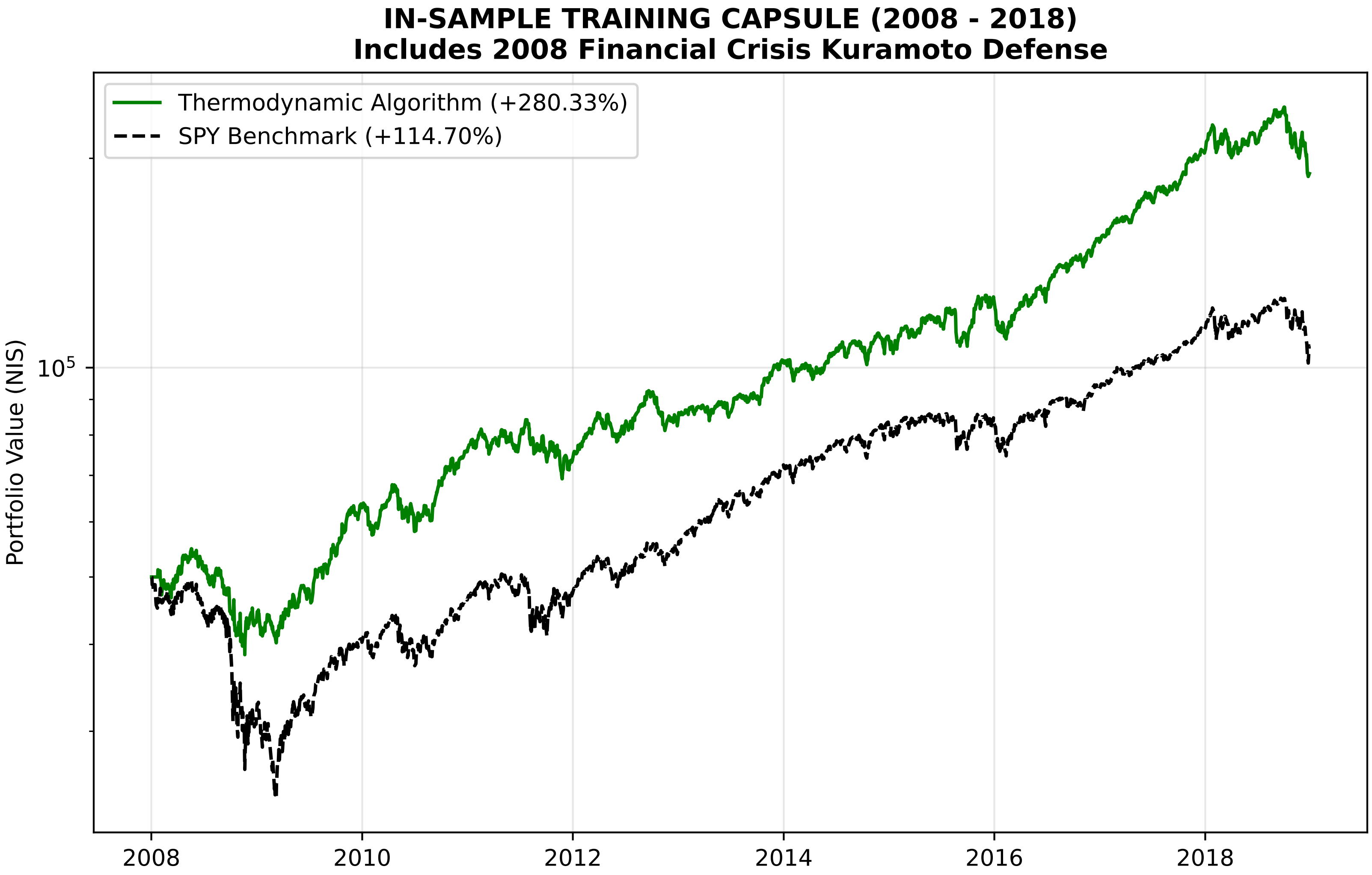
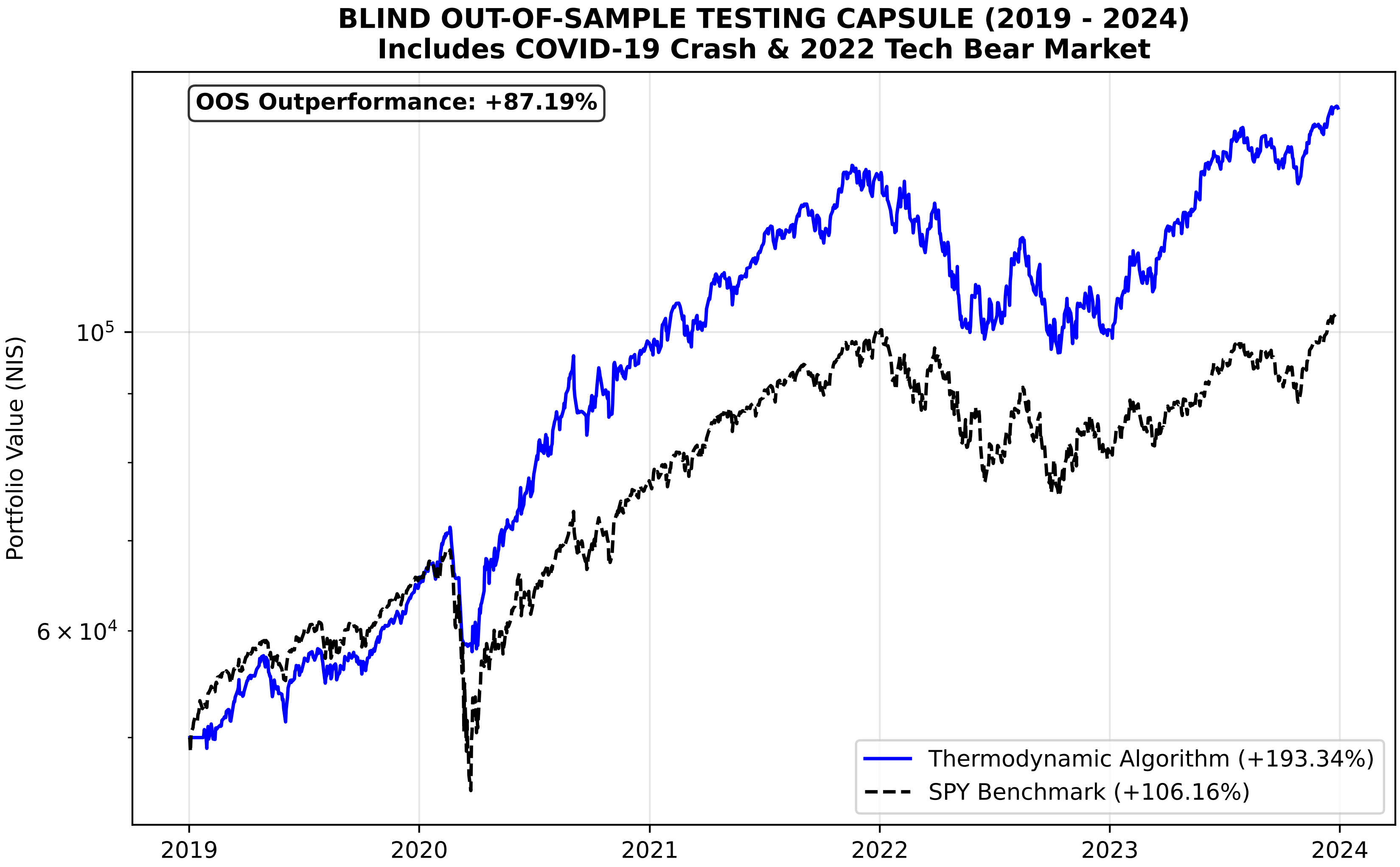

IN-SAMPLE TRAINING CAPSULE (2008 - 2018)
Includes 2008 Financial Crisis Kuramoto Defense
Thermodynamic Algorithm (+280.33%)
SPY Benchmark (+114.70%)
Portfolio Value (NIS)
10^5
2008
2010
2012
2014
2016
2018


BLIND OUT-OF-SAMPLE TESTING CAPSULE (2019 - 2024)
Includes COVID-19 Crash & 2022 Tech Bear Market
OOS Outperformance: +87.19%
Portfolio Value (NIS)
10^5
6×10^4
2019
2020
2021
2022
2023
2024
Thermodynamic Algorithm (+193.34%)
SPY Benchmark (+106.16%)